Leukemia Cell Subtyping

Blood-Smear Classification and a Case Study in Data Leakage

Capstone Project Report

A computer-vision classifier that sorts a single white blood cell into a benign look-alike or one of three stages of B-cell acute lymphoblastic leukemia — and a case study in why a 99.8% test score can be the least trustworthy figure in the portfolio.

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1. Summary

This project classifies a photograph of a single white blood cell into four classes: a benign look-alike, plus three maturation stages of B-cell acute lymphoblastic leukemia (ALL) — Early Pre-B, Pre-B, and Pro-B. The model attains **99.8% accuracy** and a **0.998 macro-F1** on the held-out test set.

> That figure is presented not as an achievement but as a **warning**. The public dataset carries no patient identifiers, so a random split almost certainly places cells from the same patient on both sides of the partition. The model may be scoring highly by recognizing per-slide staining and background rather than genuine features of leukemia.

2. Motivation

This experiment tests whether a computer-vision classifier can distinguish benign look-alike cells from leukemic blasts and separate three labeled maturation stages. Patient leakage and differences between laboratories limit what the test results establish.

3. Dataset

The project uses a public peripheral-blood-smear dataset (Aria et al., 2021) of **3,256 images**, each showing a single cell across the four classes, split 70/15/15 into training / validation / test.

A limitation that cannot be fully remedied

The dataset provides **no patient identifiers** (file names are plain indices), so there is no way to guarantee that one patient's cells remain together. Some degree of data leakage is therefore almost certainly baked into any split of this dataset, and reporting the headline figure without this caveat would be misleading.

4. Model and training

The classifier is the same architecture used for MoleCheck — **YOLO11s-cls** (5.4M parameters), transfer-learned on the blood-cell images at 224×224. Validation accuracy reached **100%** within roughly 30 epochs, which — rather than being reassuring — is itself a signal to examine the data split more critically.

5. Results

Metric	Value
Test accuracy	99.8%
Macro-F1	0.998
Errors	1 of 489 images

On most projects this would be the headline. Here it is the red flag. Cells from a single slide share staining, lighting, focus, and background, and a high-capacity model can score near-perfectly by

keying on **those** cues rather than the cell itself. This is the same failure mode the EEG project avoids through subject-level splitting; it cannot be avoided here because the identifiers are simply absent.

6. Deployment

The model was exported to ONNX for the browser demonstration and to TensorFlow Lite for the Android application (Cell Explorer), both reproducing the original model's predictions. The user can select a real cell image and classify it on-device.

7. Limitations

- Not a diagnostic tool — real leukemia diagnosis requires bone-marrow analysis, flow cytometry, and genetic testing.
- The 99.8% figure is very likely inflated by the data-leakage issue above, so it should be read as a best case, not a realistic estimate.
- A single dataset; real clinical practice presents far greater variability.

8. Future work (v2)

The next iteration moves to the **C-NMC 2019** dataset, which provides patient identifiers for roughly fifteen thousand cells. This enables holding out entire patients — as done in the EEG project — and measuring what the model has learned. The expected outcome is a **lower** accuracy than 99.8%, and that lower figure will be substantially more meaningful.